

E-Commerce Sales & Customer Behavior Report

Executive Summary

This report analyzes recent customer order data across three core categories: **Electronics**, **Clothing**, and **Home Essentials**. By evaluating purchasing patterns, customer spending segments, and cross-category behavior, this analysis aims to provide actionable business insights to drive targeted marketing and inventory management strategies.

Total gross revenue across all processed orders stands at **\$26,410.00**.

1. Customer Classification Analysis

Customers have been segmented into behavioral tiers based on their total spending. Smart threshold adjustments were applied to better reflect the dataset’s premium product pricing:

- **High-Value Buyer:** Spending > \$4,000
- **Moderate Buyer:** Spending between \$1,000 and \$4,000
- **Low-Value Buyer:** Spending < \$1,000

Customer Spend & Tier Breakdown

Customer	Total Spend	Classification Tier	Key Items Purchased
Charlie	\$6,700.00	High-Value Buyer	Phone, Chair, Cabinet, Desk
Bob	\$5,930.00	High-Value Buyer	Phone, Shirt, Tablet, Laptop
Frank	\$5,000.00	High-Value Buyer	Desk (x2)
Eve	\$4,500.00	High-Value Buyer	Tablet, Laptop

Alice	\$3,375.00	Moderate Buyer	Laptop, Trousers (x5)
Grace	\$600.00	Low-Value Buyer	Chair (x4)
David	\$305.00	Low-Value Buyer	Trousers, Shirt

2. Category Performance

The product offerings span three unique categories. Electronics serves as the massive revenue engine for the store, followed by Home Essentials.

Revenue Breakdown by Product Category

Category	Total Revenue	Revenue Share (%)	Unique Products Sold
Electronics	\$16,050.00	60.77%	Laptop, Phone, Tablet
Home Essentials	\$9,600.00	36.35%	Desk, Chair, Cabinet
Clothing	\$760.00	2.88%	Shirt, Trousers
Total	\$26,410.00	100.00%	8 Unique Items

3. Key Insights & Purchase Behavior

Premium Tech Dominance

Electronics single-handedly carries over **60% of total revenue**. This is heavily driven by high-ticket items like the Laptop (\$3,000) and Tablet (\$1,500). Even though the quantity of individual items sold in this category isn't the highest, its financial impact is unmatched.

High-Value Concentration

A striking **57% of the customer base** (4 out of 7 customers) falls into the "High-Value Buyer" category. Charlie, Bob, and Frank represent the **Top 3 Spenders**, collectively bringing in \$17,630.00—or roughly **66.7% of all company sales**.

Cross-Category Behavior (The Tech & Fashion Cohort)

Using set operations to track cross-category exploration, a specific subset of customers was identified:

- **Multi-category Engagement:** **Bob** and **Alice** are the only customers demonstrating diversified buying habits, cross-shopping between both the **Electronics** and **Clothing** sections.
- **Niche Shoppers:** Customers in the Home Essentials category (Frank and Grace) showed a strict preference for furniture, completely ignoring tech and clothing.

4. Strategic Recommendations

1. VIP Retargeting Campaigns

Because two-thirds of the revenue rests on just three customers (Charlie, Bob, Frank), implementing a high-touch VIP loyalty program is essential to maintain customer retention.

2. Cross-Selling Tech with Apparel

Since **Bob** and **Alice** proved there is an overlap between tech buyers and clothing buyers, try bundle packaging. For example, market tech accessories or sleek clothing items alongside Laptop or Tablet purchases.

3. Reposition the Clothing Line

Generating under 3% of overall revenue, clothing items (Shirts/Trousers) are currently underperforming or being treated as low-cost add-ons. Consider raising

the profile of the clothing catalog or shifting marketing budgets toward high-margin Home Essentials and Electronics.

Output: customer_orders.py

--Data--

Products: [{ 'item': 'Laptop', 'price': 3000, 'category': 'electronics'}, { 'item': 'Shirt', 'price': 40, 'category': 'clothing'}, { 'item': 'Phone', 'price': 1350, 'category': 'electronics'}, { 'item': 'Trousers', 'price': 75, 'category': 'clothing'}, { 'item': 'Tablet', 'price': 1500, 'category': 'electronics'}, { 'item': 'Cabinet', 'price': 300, 'category': 'home essentials'}, { 'item': 'Chair', 'price': 150, 'category': 'home essentials'}, { 'item': 'Desk', 'price': 2500, 'category': 'home essentials'}]

Customers: [{ 'name': 'Alice', 'age': 30, 'city': 'New York'}, { 'name': 'Bob', 'age': 25, 'city': 'Los Angeles'}, { 'name': 'Charlie', 'age': 35, 'city': 'Chicago'}, { 'name': 'David', 'age': 28, 'city': 'Houston'}, { 'name': 'Eve', 'age': 22, 'city': 'Phoenix'}, { 'name': 'Frank', 'age': 40, 'city': 'Philadelphia'}, { 'name': 'Grace', 'age': 27, 'city': 'San Antonio'}]

Orders: [{ 'customer': 'Alice', 'product': 'Laptop', 'quantity': 1}, { 'customer': 'Alice', 'product': 'Trousers', 'quantity': 5}, { 'customer': 'Bob', 'product': 'Phone', 'quantity': 1}, { 'customer': 'Bob', 'product': 'Shirt', 'quantity': 2}, { 'customer': 'Bob', 'product': 'Tablet', 'quantity': 1}, { 'customer': 'Bob', 'product': 'Laptop', 'quantity': 1}, { 'customer': 'Charlie', 'product': 'Phone', 'quantity': 2}, { 'customer': 'Charlie', 'product': 'Chair', 'quantity': 6}, { 'customer': 'Charlie', 'product': 'Cabinet', 'quantity': 2}, { 'customer': 'Charlie', 'product': 'Desk', 'quantity': 1}, { 'customer': 'David', 'product': 'Trousers', 'quantity': 3}, { 'customer': 'David', 'product': 'Shirt', 'quantity': 2}, { 'customer': 'Eve', 'product': 'Tablet', 'quantity': 1}, { 'customer': 'Eve', 'product': 'Laptop', 'quantity': 1}, { 'customer': 'Frank', 'product': 'Desk', 'quantity': 2}, { 'customer': 'Grace', 'product': 'Chair', 'quantity': 4}]

--Tasks--

1. Store customer orders

['Alice', 'Bob', 'Charlie', 'David', 'Eve', 'Frank', 'Grace']

Customer names stored in a list: ['Alice', 'Bob', 'Charlie', 'David', 'Eve', 'Frank', 'Grace']

[('Alice', 'Laptop', 3000, 'electronics', 1), ('Alice', 'Trousers', 75, 'clothing', 5), ('Bob', 'Phone', 1350, 'electronics', 1), ('Bob', 'Shirt', 40, 'clothing', 2), ('Bob', 'Tablet', 1500, 'electronics', 1), ('Bob', 'Laptop', 3000, 'electronics', 1), ('Charlie', 'Phone', 1350, 'electronics', 2), ('Charlie', 'Chair', 150, 'home essentials', 6), ('Charlie', 'Cabinet', 300, 'home essentials', 2), ('Charlie', 'Desk', 2500, 'home essentials', 1), ('David', 'Trousers', 75, 'clothing', 3), ('David', 'Shirt', 40, 'clothing', 2), ('Eve', 'Tablet', 1500, 'electronics', 1), ('Eve', 'Laptop', 3000, 'electronics', 1), ('Frank', 'Desk', 2500, 'home essentials', 2), ('Grace', 'Chair', 150, 'home essentials', 4)]

Generate a dictionary where keys are customer names and values are lists of ordered products
{ 'Alice': ['Laptop', 'Trousers'], 'Bob': ['Phone', 'Shirt', 'Tablet', 'Laptop'], 'Charlie': ['Phone', 'Chair', 'Cabinet', 'Desk'], 'David': ['Trousers', 'Shirt'], 'Eve': ['Tablet', 'Laptop'], 'Frank': ['Desk'], 'Grace': ['Chair']}

2. Classify products by category

{ 'Laptop': 'electronics', 'Shirt': 'clothing', 'Phone': 'electronics', 'Trousers': 'clothing', 'Tablet': 'electronics', 'Cabinet': 'home essentials', 'Chair': 'home essentials', 'Desk': 'home essentials'}
{ 'home essentials', 'clothing', 'electronics'}

3. Analyze customer orders

{ 'Alice': 3375, 'Bob': 5930, 'Charlie': 6700, 'David': 305, 'Eve': 4500, 'Frank': 5000, 'Grace': 600}

```
{'Alice': 'moderate buyer', 'Bob': 'high-value buyer', 'Charlie': 'high-value buyer', 'David':  
'low-value buyer', 'Eve': 'high-value buyer', 'Frank': 'high-value buyer', 'Grace': 'low-value buyer'}
```

4. Generate business insights

```
{'electronics': 16050, 'clothing': 760, 'home essentials': 9600}  
{'Trousers', 'Shirt', 'Desk', 'Tablet', 'Cabinet', 'Laptop', 'Phone', 'Chair'}  
['Alice', 'Bob', 'Bob', 'Bob', 'Charlie', 'Eve', 'Eve']  
[(('Charlie', 6700), ('Bob', 5930), ('Frank', 5000))]
```

5. Organize and display data

Alice spent a total of \$3375.00 and is classified as a moderate buyer.

Bob spent a total of \$5930.00 and is classified as a high-value buyer.

Charlie spent a total of \$6700.00 and is classified as a high-value buyer.

David spent a total of \$305.00 and is classified as a low-value buyer.

Eve spent a total of \$4500.00 and is classified as a high-value buyer.

Frank spent a total of \$5000.00 and is classified as a high-value buyer.

Grace spent a total of \$600.00 and is classified as a low-value buyer.

```
{'electronics': {'Bob', 'Charlie', 'Eve', 'Alice'}, 'clothing': {'Bob', 'Alice', 'David'}, 'home essentials':  
{'Grace', 'Charlie', 'Frank'}}  
{'Bob', 'Alice'}
```

--Actions--

Customer order processing in python

```
[('Alice', 'Laptop', 3000, 'electronics', 1), ('Alice', 'Trousers', 75, 'clothing', 5), ('Bob', 'Phone',  
1350, 'electronics', 1), ('Bob', 'Shirt', 40, 'clothing', 2), ('Bob', 'Tablet', 1500, 'electronics', 1),  
(('Bob', 'Laptop', 3000, 'electronics', 1), ('Charlie', 'Phone', 1350, 'electronics', 2), ('Charlie',  
'Chair', 150, 'home essentials', 6), ('Charlie', 'Cabinet', 300, 'home essentials', 2), ('Charlie',  
'Desk', 2500, 'home essentials', 1), ('David', 'Trousers', 75, 'clothing', 3), ('David', 'Shirt', 40,  
'clothing', 2), ('Eve', 'Tablet', 1500, 'electronics', 1), ('Eve', 'Laptop', 3000, 'electronics', 1),  
(('Frank', 'Desk', 2500, 'home essentials', 2), ('Grace', 'Chair', 150, 'home essentials', 4))]  
{'name': 'Alice', 'age': 30, 'city': 'Boston'}
```

Classification and analysis:

```
{'Alice': 'moderate buyer', 'Bob': 'high-value buyer', 'Charlie': 'high-value buyer', 'David':  
'low-value buyer', 'Eve': 'high-value buyer', 'Frank': 'high-value buyer', 'Grace': 'low-value buyer'}  
{'electronics': {'Tablet', 'Laptop', 'Phone'}, 'clothing': {'Shirt', 'Trousers'}, 'home essentials':  
{'Chair', 'Desk', 'Cabinet'}}
```